IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Dense Prediction

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Dense Prediction

01 **Spatial Outputs**

Choose the output granularity

02 **Grounding Text**

Link phrases to boxes

03 **Prompting Masks**

Refine boxes into pixels

01 · Spatial Outputs

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Refine boxes into pixels

Why Keep Location

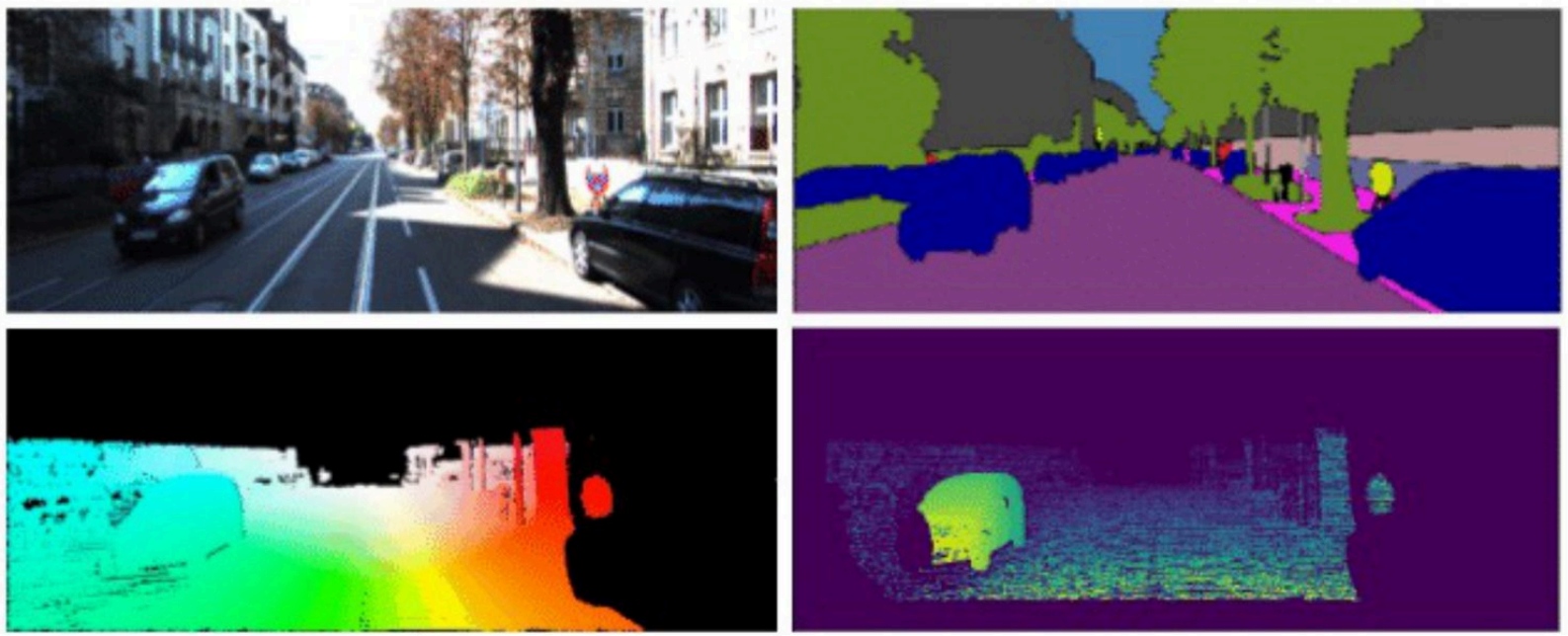
A class label cannot specify where the object is

Image answer	dog	?	?
Required output	class	box	mask

A class name leaves the object location unspecified

Dense Visual Tasks

Different tasks preserve different spatial targets



optical flow

depth

Pixel outputs can represent category, motion or depth

Output Granularity

Image, box and pixel outputs answer different questions

Output	Question	Shape
Image label	What is present?	C
Boxes	Where are instances?	N × 4
Mask	What occupies each pixel?	H × W

Spatial Feature Grid

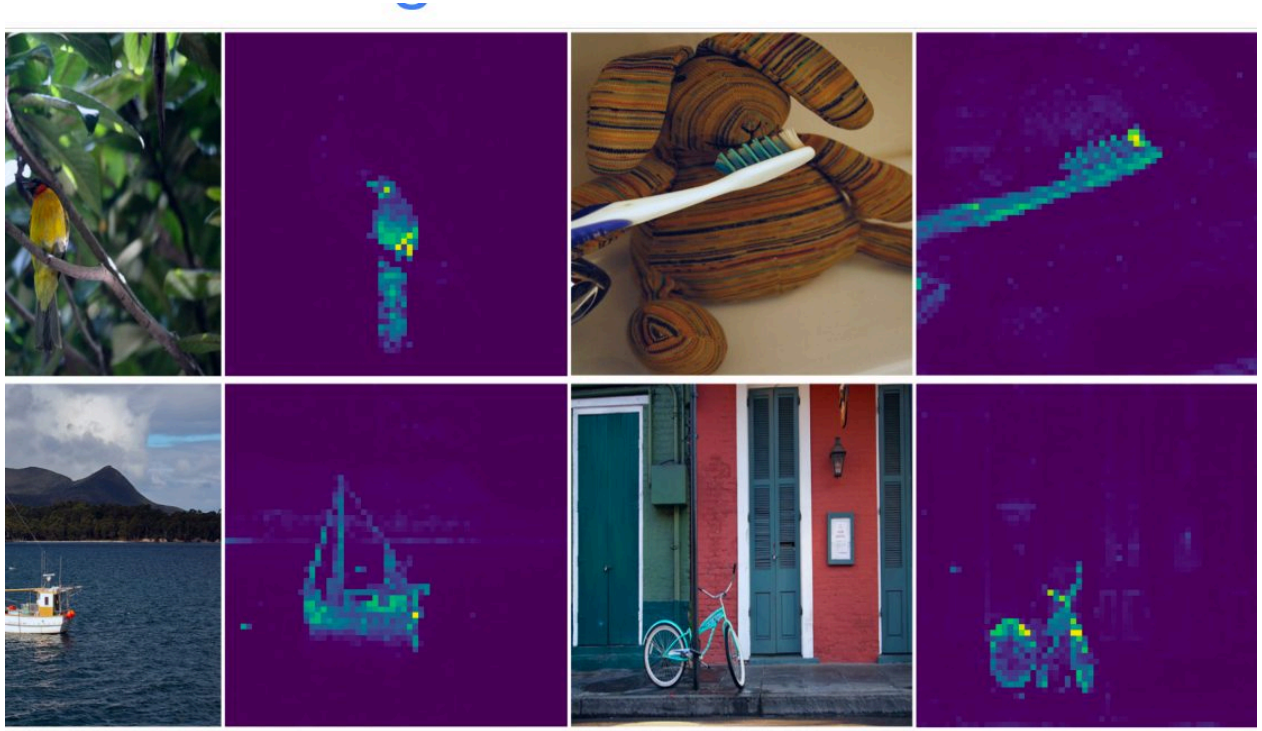
Patch embeddings retain a grid before pooling

row 1	z11	z12	z13
row 2	z21	z22	z23

Each feature vector has a spatial address

DINO Attention Maps

Spatial attention can highlight objects without mask labels



Attention can reveal structure before dense supervision

https://arxiv.org/abs/2104.14294

Caveat: Attention Is Not a Mask

A heatmap does not define a calibrated pixel classifier

row 1	0.1	0.7	0.9
row 2	0.2	0.6	0.4
row 3	0.05	0.1	0.2

Thresholding changes the selected region

Bounding Box Coordinates

Coordinate conventions must match the image scale

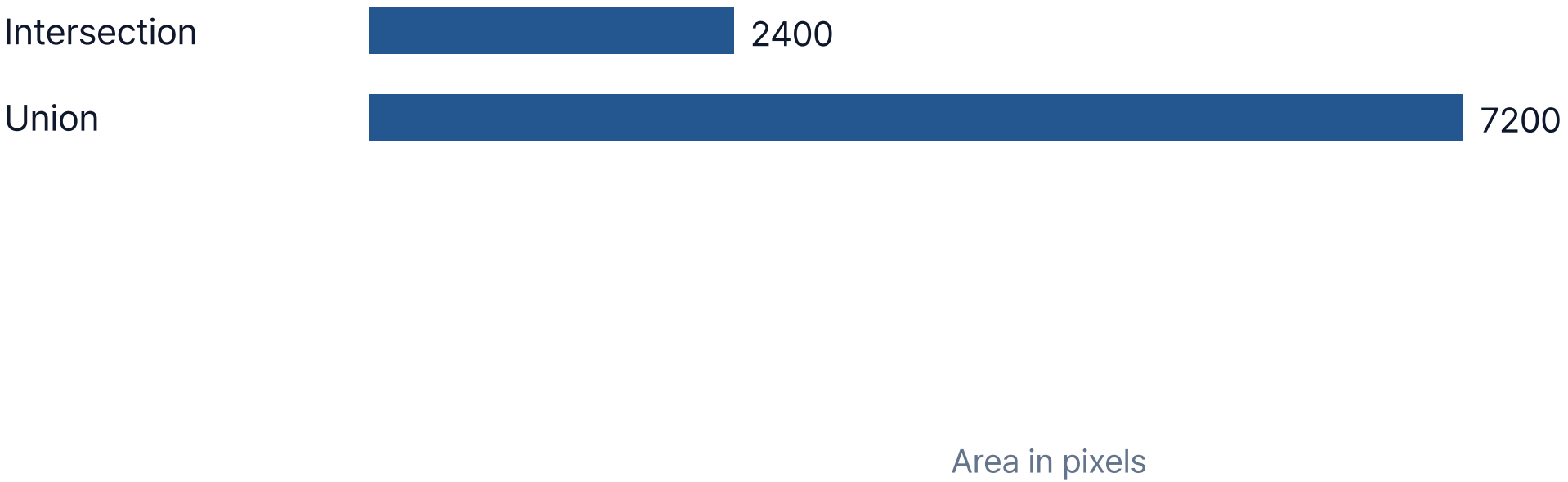
	x1	y1	x2	y2
Pixels	20	30	70	90
Normalized	0.2	0.3	0.7	0.9

One box has several valid coordinate conventions

Intersection over Union

Overlap is measured relative to the union

$$IoU(A, B) = |A \cap B| / |A \cup B|$$



Demo: Box Overlap

Move a predicted box and inspect intersection over union

Offset 0



Two equal boxes; horizontal displacement only

Offset 0 | intersection 4800 | union 4800 | IoU 1.000
IoU = intersection area / union area

Answer: Overlap Arithmetic

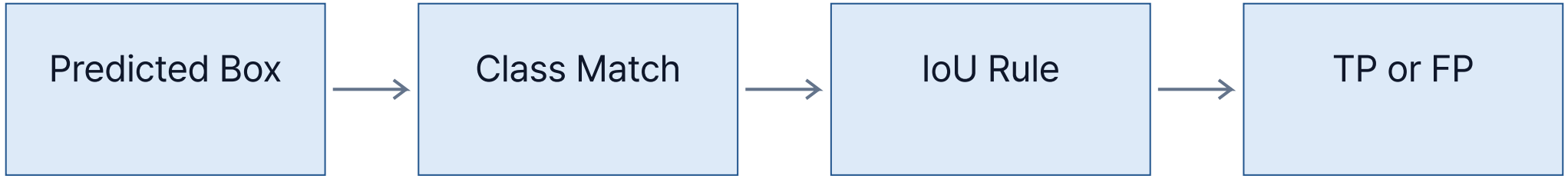
Two equal rectangles can share little area

	Width	Height	Shift	A∩B	A∪B
Offset 40	80	60	40	2400	7200

A forty-pixel shift gives intersection over union of one third

Detection Evaluation

A match requires both location and class agreement



Location alone does not make a correct detection

02 · Grounding Text

01 Spatial Outputs

Choose the output granularity

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Refine boxes into pixels

Why Open Vocabulary

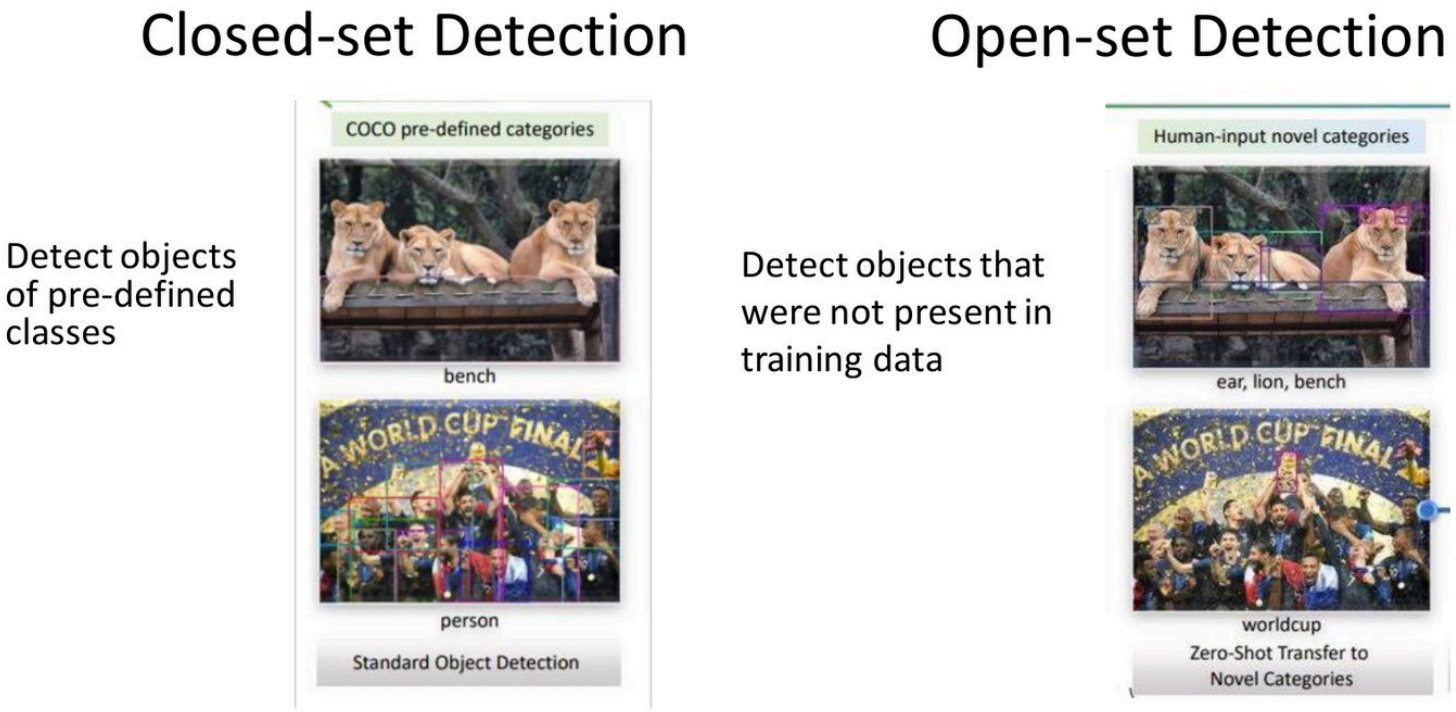
A fixed label set cannot express every requested object

Item	Field 1	Field 2	Field 3
Fixed labels	cat	dog	car
User phrases	red toolbox	damaged handle	?

A user phrase may fall outside the training label list

Closed and Open Sets

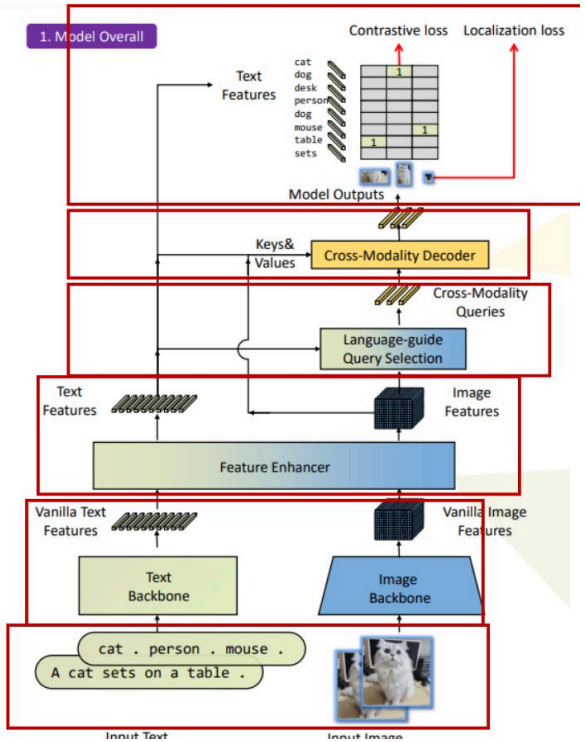
Text expands the set of detection queries



Detection queries can be supplied as text

Grounding DINO

Language and image features meet before box prediction



Text conditions which boxes are proposed

https://arxiv.org/abs/2303.05499

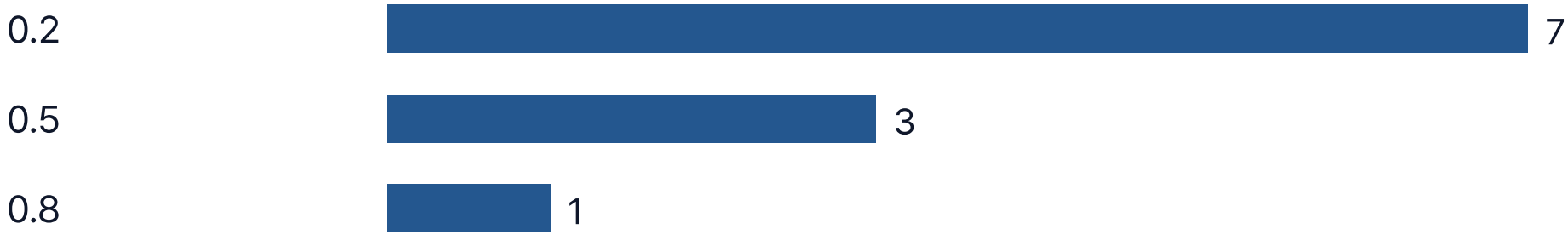
Phrase Grounding Output

A region must remain attached to its query phrase

Phrase	Box	Score
red valve	(10,20,40,60)	0.72
blue pump	(50,15,90,65)	0.61

Confidence Thresholds

A score threshold changes retained boxes



Retained boxes in an illustrative example

Raising the threshold removes more candidates

Demo: Grounding DINO

Text prompts produce measured boxes and confidence scores

Code | Results

Run in Colab ↗

```
image = Image.open(image_paths[0]).convert("RGB")
boxes, scores, labels = detect_objects(image, ["dog", "cat"], box_threshold=0.30)
draw_detections(image, boxes, scores, labels)
plt.show()
```

Change the recorded prompt or threshold. Which objects disappear first?

Model scores are not calibrated probabilities

https://huggingface.co/IDEA-Research/grounding-dino-tiny

Suppression and Duplicate Boxes

Several boxes may describe the same object

	x1	y1	x2	y2
Box A	10	10	40	40
Box B	11	12	42	39
Box C	70	10	90	30

Duplicate handling is part of the detector pipeline

Worked Example: Greedy NMS

Trace the inputs before execution

```
Box 0: [0,0,80,60], score .9
Box 1: [10,0,90,60], score .8
Box 2: [120,0,180,60], score .7; IoU threshold .5
```

Demo: Greedy NMS

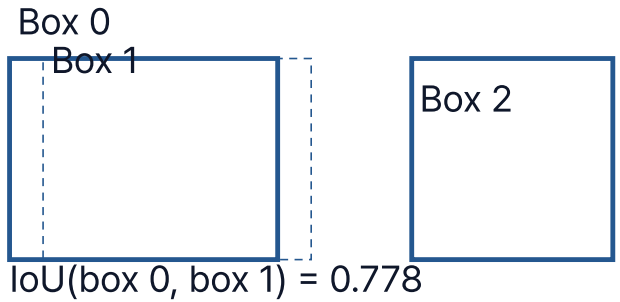
Executable educational calculation; no pretrained inference

Compare IoU thresholds 0.1 and 0.9; then change which box has the highest score

Boxes: x1 y1 ×2 y2 score

```
0 0 80 60 .9
10 0 90 60 .8
120 0 180 60 .7
```

Run NMS | IoU threshold 0.5



Box 0: 0.9 → KEEP

Box 1: 0.8 → SUPPRESS

Box 2: 0.7 → KEEP

Kept IDs: 0, 2; suppressed: 1.

Questions: Which score wins? When do two real objects suppress each other?

Answer: Greedy NMS

Same inputs as the initial demo state

```
IoU(0,1)=.777778
keep IDs [0,2], suppress [1]
```

Failure: Greedy NMS

Identify the assumption that breaks

```
Single-class NMS can suppress two real overlapping objects
scores must already exist, NMS is not detection
```

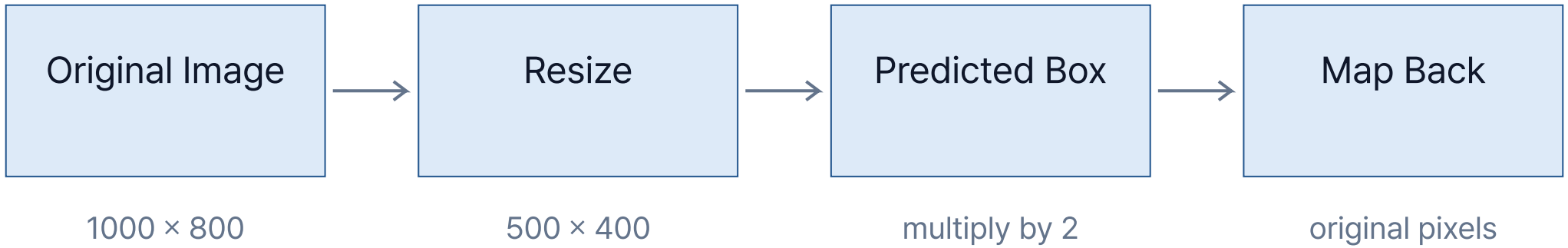
Prompt Ambiguity

Relational phrases can require reasoning beyond object identity

Phrase	Ambiguity
the valve	Several valves
left valve	Camera viewpoint
broken valve	Unseen defect concept

Coordinate Rescaling

Resize operations must be inverted consistently



Undo the exact geometric preprocessing

Crop Transform Code

Store scale and offset with each crop

```
xyxy = np.array([10., 20., 40., 60.])
scale = np.array([2., 2., 2., 2.])
offset = np.array([100., 50., 100., 50.])
original_xyxy = xyxy * scale + offset
```

Grounding Failure Slices

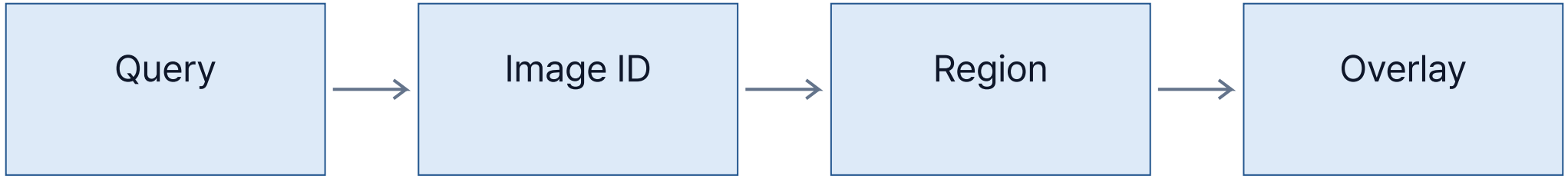
Inspect small, occluded and rare-query objects separately

	Found	Missed
Large	8	2
Small	5	5
Occluded	3	7

The same total score can hide different failure slices

Grounding Contract

Return both the region and the evidence needed to verify it



A verifiable region retains its original image

03 · Prompting Masks

01 Spatial Outputs

Choose the output granularity

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03 **Prompting Masks**

Refine boxes into pixels

Why Promptable Masks

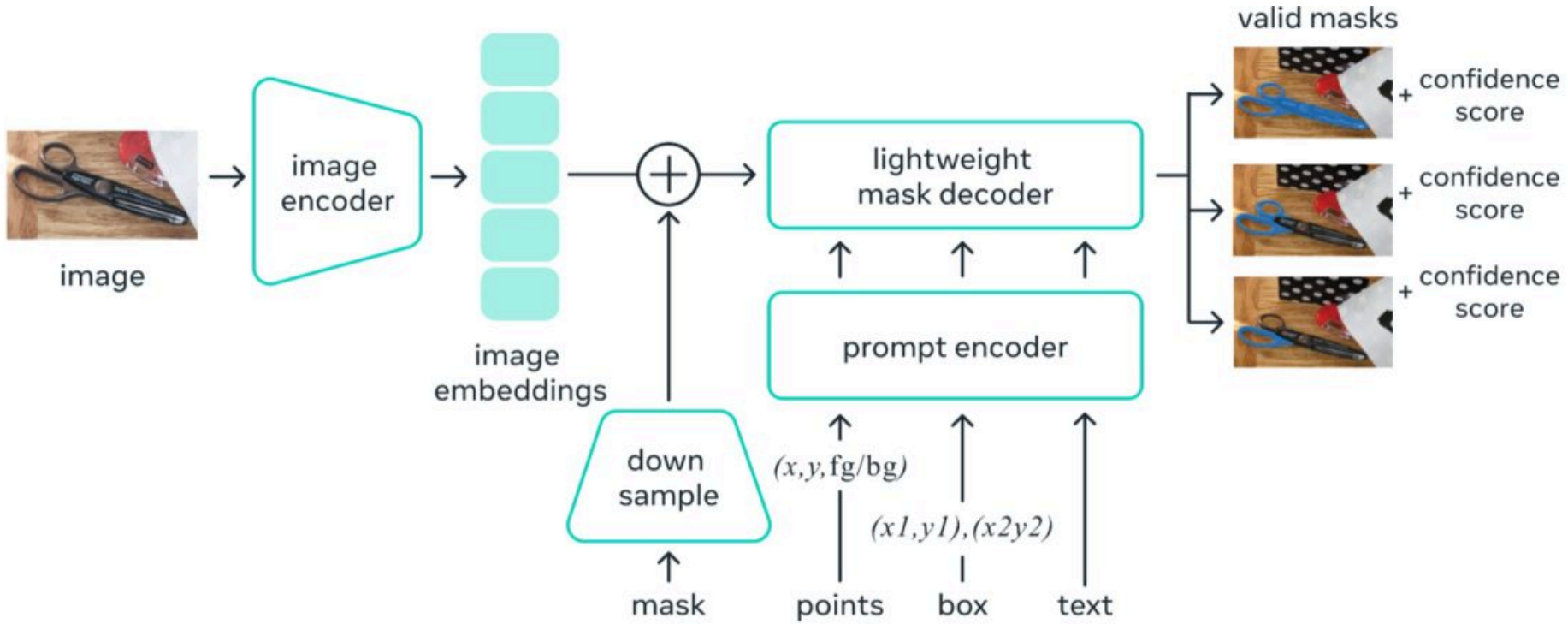
A box still includes background pixels

r1	0	0	0	0
r2	0	1	1	0
r3	0	1	0	0
r4	0	0	0	0

A mask separates object pixels from box background

Segment Anything

An image embedding is reused across prompts



Prompt changes can reuse the image embedding

Mask Prompt Examples

Points and boxes specify the intended object



The prompt identifies the desired region

Worked Example: Click to Mask

Trace the inputs before execution

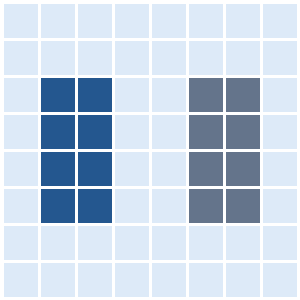
```
8×8 grid; two separate objects have intensity 8
Background intensity 1; target is the left object
Seed (row 2, col 1), tolerance 0; use four-neighbor flood fill
```

Demo: Click to Mask

Executable educational calculation; no pretrained inference

Click each object and the background; raise tolerance to 7 and inspect mask leakage

Intensity tolerance 0



Click a cell; seed = (2, 1)

Region pixels = 8; IoU = 1.000

Dice = 1.000; target = left object

Intensity flood fill using tolerance, not SAM.
Questions: Which object does a click select? What changes at tolerance 7?

Answer: Click to Mask

Same inputs as the initial demo state

```
Selected area = 8 pixels
target IoU = 1 and Dice = 1
```

Failure: Click to Mask

Identify the assumption that breaks

```
Intensity connectivity does not understand object identity
this flood fill illustrates prompting and is not SAM inference
```

Demo: SAM Prompt Ambiguity

Point and box prompts on real pet images

Code | Results

Run in Colab ↗

```
image = Image.open(image_paths[1]).convert("RGB")
point = [[image.width // 2, image.height // 2]]
mask, predicted_quality = segment_with_points(image, point)
plt.imshow(image); plt.imshow(mask, alpha=0.4)
plt.show()
print("Predicted mask quality:", predicted_quality)
```

Which region does the point select? Compare the three candidate masks.

SAM predicts mask quality; the displayed score is not measured ground-truth IoU

Mask Metrics

Class imbalance makes pixel accuracy misleading

Actual / Predicted	Background	Object
Background	96	1
Object	2	1

Background pixels can dominate accuracy

Dice and IoU

Both metrics compare predicted and target regions

$$\text{Dice} = 2|A \cap B| \,/\, (|A| + |B|)$$

Grounded Segmentation

Text proposes a box and a mask model refines its pixels



A box proposal connects text to a mask

https://arxiv.org/abs/2303.05499

Error Propagation

The mask stage cannot repair every missed proposal



Illustrative object counts

A missed proposal removes the downstream opportunity

Demo: Grounded Segmentation

Grounding DINO boxes become SAM prompts

Code | Results

Run in Colab ↗

```
image = Image.open(image_paths[0]).convert("RGB")
boxes, scores, labels, masks, quality = grounded_sam(image, ["dog"],
box_threshold=0.30)
visualize_grounded_sam(image, boxes, scores, labels, masks)
print("Detected boxes:", len(boxes), "Masks:", len(masks))
```

Inspect a failed box. Can the mask stage repair the wrong object selection?

The displayed masks come from actual Grounding DINO + SAM inference

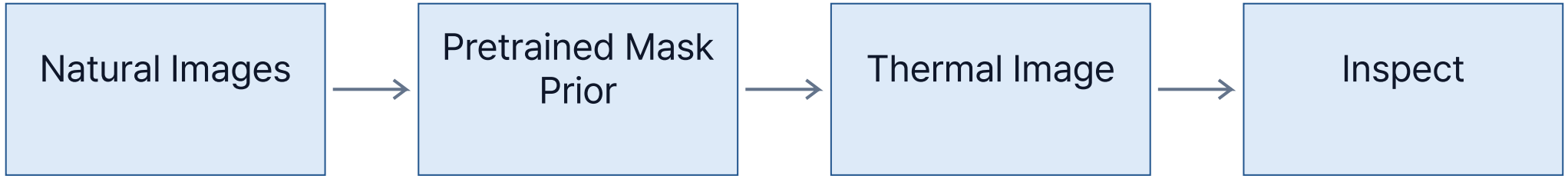
Mask Area Calculation

Pixel counts need the original spatial scale

```
mask = np.array([[0,1,1],[0,1,0]], dtype=bool)
area_pixels = int(mask.sum())
pixel_area_mm2 = 0.25
area_mm2 = area_pixels * pixel_area_mm2
```

Caveat: Domain Transfer

A natural-image mask prior may fail on a new sensor



A new sensor changes the image distribution

Dense Prediction Audit

Verify coordinates, matching and mask boundaries together

Layer	Check
Geometry	Overlay in original pixels
Detection	Class and IoU match
Segmentation	Boundary and object IoU

Takeaway: Dense Prediction

01 Keep spatial coordinates with every prediction

02 Open vocabulary changes queries, not verification duties

03 Audit proposal errors before interpreting masks